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# **Reliability and Validity of Using EQ-5D-5L among Healthy and Adolescents with Major Mental Health Disorders in Ethiopia**

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## **Authors' contributions**

AG, GBG, and YB conceived the idea; AG, ES, CM and YB designed the study, acquired the funding, and provided detailed information regarding data collection processes in Ethiopia. AG and GBG performed data quality control. AG and MK prepared the draft manuscript. All authors reviewed the analysis, interpretation of the results, and the final manuscript.

### **Declaration of Interest**

Elly Stolk is a member of the EuroQol Research Foundation. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

### **Ethical considerations**

Ethical approval was secured from Ethics Review Committee of School of Pharmacy, Addis Ababa University, Ethiopia and prior permission was sought from Demographic Health Surveillance site of Butajira Rural Health Project (BRHP) office, Ethiopia. Written Informed consent was obtained from all study participants to confirm their willingness for participation after explaining the purpose of study.

## **Abstract**

**Background:** The EQ-5D is a validated and widely used generic measure of health-related quality of life (HRQoL) in both healthy individuals and those with various medical conditions. The objective of this study was to test whether EQ-5D-5L is reliable and valid for use among school sample adolescents and those with major mental health disorders in Ethiopia.

**Methods** Participants were recruited from ten sub-districts comprising the Butajira Rural Health Programme (BRHP) and Butajira major mental health disorders center. Data were collected using an Amharic (Ethiopia) EQ-5D-5L self-complete -paper and the questionnaire was administered 10 days after the first completion for test-retest procedures. Two-way mixed-effects models absolute intraclass correlation coefficient (ICC) was used to test reliability of the instrument while Kruskal-Wallis rank test with pairwise comparison was used to assess the known group validity of the instrument.

**Results:** There were 501 (201 school sample and 300 adolescents with major mental health disorders) participants recruited and 497 were included in the sample for analysis. The ICC was high ( $ICC > 0.7, p < 0.001$ ) for all EQ-5D-5L dimensions, EQ-5D-5L utility index and EQ-VAS scores. The findings revealed that the Amharic EQ-5D-5L has significant known group validity as shown by the difference in scores among various disease group (depression, schizophrenia, and bipolar) and experience of chronic illness.

**Conclusions:** The results shows that the Amharic EQ-5D-5L is reliable and valid instrument for the measurement of HRQoL among adolescent populations in Ethiopia.

**Keywords:** Adolescent, EQ-5D-5L, Ethiopia, HRQoL, Mental health, Quality of Life

## Introduction

The prevalence of childhood and adolescent emotional and behavioral disorders in Ethiopia is approximately 17% (1). Several strategies have been established to address this disease burden, with one of the strategic priorities being to ensure the affordability and accessibility of psychotropic drugs (2–4). In order to adequately understand the burden of mental health conditions, and evaluate the effectiveness and cost-effectiveness of treatments, it is important to measure health outcomes in a valid and reliable way. The health of the population cannot be well characterized from the analysis of mortality and morbidity statistics alone. There is also a need to characterize health in terms of broader dimensions such as symptoms, function, and overall well-being (5). This broader conceptualization of health has led to increasing development and use of instruments designed to measure health-related quality of life (HRQoL). HRQoL is a multi-dimensional concept which focuses on the effect of disease or a health condition and its treatment on a person's daily life (6,7). HRQoL measures can be preference-based, based on the valuation of health outcomes, or psychometric, based on domain specific symptom and function scores. Preference-based measures generate a health utility value on a scale anchored at 0 (dead) and 1 (full health), with some measures having values that are less than 0. These utility values can be combined with length of life to generate quality-adjusted life years (QALYs) estimates for cost-utility assessment (8,9).

One of the most commonly used generic preference-based measures of utility in adults is the EQ-5D which was developed by a multi-disciplinary group of researchers in European countries, but has now been translated into more than 130 languages with various modes of administration. The EQ-5D has a visual analogue scale (EQ-VAS) on a 100 (best imaginable health) to 0 (worst imaginable health) scale and five question related to mobility, self-care, usual activities,

pain/discomfort and anxiety/depression which can be scored using utility values that represent population preferences. The five questions initially included 3-level (EQ-5D-3L) response options (no problems, some/moderate problems, extreme problems/unable to/confined to bed). Subsequently, the EuroQol Group established the five-level (EQ-5D-5L) response option version in 2009 to improve the instrument's sensitivity and eliminate ceiling effects (10). The EQ-5D-5L has been used to evaluate general health, long-term disability, medical conditions and psychosocial treatment. This input is of particular value in resource allocation decision making (11,12). The EQ-5D-5L has been translated to Amharic (the official language of Ethiopia), using the standardized approach recommended by the EuroQol group (13). A value set has also been developed for the EQ-5D-5L from the Ethiopian general population using the EQ-PVT software of EQ-5D-5L valuation study (14), demonstrating that it is feasible and culturally acceptable to measure preferences for health outcomes in Ethiopia.

The HRQoL of children and adolescents is attracting more attention within health care systems. One measure of this is the increasing number of published studies evaluating the feasibility, reliability, and validity of different health status measurement tools in children and adolescents (13,15,16). The EQ-5D-3L and EQ-5D-5L were originally developed for use with adults. A child and adolescent version were developed in 2009 – the EQ-5D-Y (Youth, 3 levels of severity), using age-appropriate modification of questions, for example, anxiety/depression is expressed as feeling worried, sad or unhappy (17,18). The EQ-5D-Y has been shown to be valid for use in children and adolescents in several countries including Germany, Italy, South Africa, Spain, and Sweden (15,16) and is recommended for use for those aged 8 to 15 years (10). Although different countries use the EQ-5D-Y in children and adolescents for measuring health outcomes in clinical and general population settings, the EQ-5D-Y user guide suggests that either the EQ-5D-Y or adult EQ-5D versions can be

used for individuals aged 12-15 years (17). The use of the EQ-5D-3L was also found to be adequate in those aged 12 to 18 years although concerns about the content validity prompted the development of the EQ-5D-Y(19). Neither EQ-5D nor EQ-5D-Y have been applied in an adolescent population in the Ethiopian setting. However, before using the tool in a low-income country like Ethiopia, it should be evaluated as differences in communication style, language, culture and values may be associated with differences in the conceptualization of health. EQ-5D-Y does not yet have utility values that can be used in economic evaluation but the EQ-5D-5L Amharic version has utility values (range: -0.718 to 1) (14). Therefore, this study aims to investigate the measurement properties, reliability and validity of the adult version of EQ-5D-5L in a school sample adolescent population as well as adolescents with mental health conditions.

## **Methods**

### **Study design**

A repeated cross-sectional study design in the same population was used to recruit participants from schools and hospitals using a face-to-face administration of the Amharic (Ethiopia) EQ-5D-5L self complete- paper. After ten days, participants were asked to complete the measure again.

### **Study setting**

This study was conducted in one urban and nine rural kebeles of Butajira District, which is part of the Butajira Demographic Surveillance Site (DSS), located in Gurage Zone, in the Southern Nations, Nationalities and Peoples Regional State of Ethiopia. These study areas were selected because they reflect the urban and rural localities of the country. The populations of the areas covered are mixed in terms of ethnicity and culture. The country has successfully implemented its policy of expanding and re-establishing primary health care facilities over the last 25 years. As a result, 16,440 health posts, 3,547 health centers, and 311 hospitals have been established (HSDP-IV, 2014/2015).

## **Sampling method and study population**

A cluster sampling strategy was used to select 201 study participants from all of the ten sub-districts in south-central Ethiopia covered by the Butajira Rural Health Programme (BRHP). Each sub-district includes one elementary and one high school. For each school (n=20), a supervisor selected one class from both elementary and high schools using simple random sampling techniques. Within each class (n=40-60), 20 students between the age of 12 and 17 were randomly selected and were invited to complete the Amharic EQ-5D-5L and EQ-VAS by trained data collectors. For the major mental health disorders (schizophrenia, bipolar disorder and depressive disorder) participants were identified by psychiatry nurses at the Butajira hospital, a major mental health disorder center. Patients who had a scheduled follow-up visit at the center and who were not experiencing an acute worsening of their condition were identified consecutively until 100 patients had been identified for each disorder (n= 300 respondents).

## **Data Collection and Fieldwork**

Data collection was undertaken in the selected schools and the hospital. Trained data collectors (n=4) explained the objective of the study and what the questionnaire was before handing out a paper-based version of the questionnaire to participants for them to self-complete. The psychiatry nurses acted as data collectors in the hospital. Prior to data collection, parents were informed of the methodology and objectives of the study by means of an official letter written by the researchers that included an informed consent form. Participants were provided the consent form signed by their parents.

The questionnaire included demographic and health items, and health measurement using the EQ-5D-5L, and the EQ-VAS scaled from 0 (the worst health you can imagine) to 100 (the best health

you can imagine). All study participants received the questionnaire again 10 days after the first completion for the test-retest procedures to investigate reliability of the instrument.

### **Quality Control (QC)**

Data collectors were trained on the concept of HRQoL, and the use of the EQ-5D-5L as a generic questionnaire to value health states. The supervisors and data collectors checked the completeness of responses, especially for the second interview.

### **Ethical approval**

Ethical approval was secured from Ethics Review Committee of School of Pharmacy, Addis Ababa University, Ethiopia and prior permission was sought from Demographic Health Surveillance site of Butajira Rural Health Project (BRHP) office, Ethiopia.

### **Data Analysis**

Descriptive statistics including mean, standard deviation (SD), percentage and percentage scoring at the floor and the ceiling were used to assess data characteristics of participants' sociodemographic data and the five separate domains of EQ-5D-5L. Ceiling effects are measured by the proportion of respondents getting the highest score, while floor effects are measured using the proportion of respondents reporting the lowest possible score.

The ability of the EQ-5D-5L to produce a consistent result at two different times (the first completion, and 10 days thereafter) was evaluated using test-retest reliability. The procedure consisted of applying the same measurement at two different times where change is assumed to not have occurred (20). We attributed change in scores over this period to random error (21,22). A two-way mixed-effects model, absolute ICC (23) was used to estimate the stability of the EQ-5D-5L and

EQ-VAS scores. This approach considers measurement errors which other correlation coefficients, such as Pearson or Spearman do not consider. An ICC > 0.7 was considered acceptable (23).

The ability of the EQ-5D-5L and EQ-VAS score to discriminate between subgroup with and without major mental health disorder, and other sociodemographic characteristics were assessed by known group or discriminant validity. The known group validity was examined in terms of several known relationships hypotheses that are found in other population-based validation studies. The first hypothesis was that the school sample and those with mental health conditions would have similar missing responses/ambiguous answers. The second hypothesis was that there would be more variation in the pain and discomfort, as well as the anxiety and depression dimensions. The third hypothesis was that participants with significant health problems (in our case adolescents with major mental health disorders) would report poorer health status than those without known mental health problems in terms of utility values and EQ-VAS scores. We compared: (a) adolescents with major mental health disorders and adolescents with no known mental health disorder; (b) females and males; (c) those with and without experience of chronic illness; and (d) adolescents with lower and higher educational status.

Patient's utility score were computed using the disutility weights of Ethiopian general population (14). Microsoft Office Excel 2016 was used to calculate the tariff EQ-5D-5L utility using the following formula: *Utility value* =  $1 - (mo2 * (0.0337) + mo3 * (0.0644) + mo4 * (0.2276) + mo5 * (0.3598) + sc2 * (0.0235) + sc3 * (0.0395) + sc4 * (0.1419) + sc5 * (0.2223) + ua2 * (0.0323) + ua3 * (0.0483) + ua4 * (0.1574) + ua5 * (0.2721) + pd2 * (0.0361) + pd3 * (0.0516) + pd4 * (0.2703) + pd5 * (0.4064) + ad2 * (0.0259) + ad3 * (0.0848) + ad4 * (0.2987) + ad5 * (0.4578))$ .

mo=mobility, sc=self-care, ua=usual activity, Pd=pain and discomfort, ad=Anxiety and depression

Comparisons and correlation between health state utility index and EQ-VAS scores of school samples and adolescents with major mental health disorder were assessed using non-parametric Kruskal-Wallis rank test with a pair-wise comparison for both utility index and VAS scores to determine the validity of the instrument. The statistical significance level was set at  $p < 0.05$ .

## **Results**

### **Respondent Characteristics**

There were 501 participants who completed the questionnaire. There were four missing responses in the EQ-5D-5L or EQ-VAS. A total of 497 formed the sample for analysis. About two-thirds of patients with major mental health disorders (59.22%) were males. The mean (SD) age was 15.57 (1.35) years and 41.80 % of the school sample adolescents had experience of chronic illness within their life time, either from their own experience, from illness in the family, or through caring for others (Table 1).

### **Self-Reported Health Problems**

Table 2. shows the proportion of respondents reporting problems on EQ-5D-5L dimensions by disease type. In all disease types, the highest proportions of health problems were reported in the anxiety/depression and pain/discomfort dimensions. The lowest proportion of health problems was in the self-care dimension for school sample adolescents and in the mobility dimension for participants with mental health disorders. The mean self-reported EQ-VAS score was 92.37 (SD=11.81), 72.12 (SD=19.15), 67.55 (SD=19.55), and 68.81 (SD=16.05) for the school sample adolescent, bipolar, schizophrenia, and depression patients, respectively. In general, the proportion of respondents reporting any problem was highest in schizophrenia and depression patients. School

sample adolescent respondents reported fewer problems compared to respondents with major mental health disorders.

### **Floor and ceiling effects**

Floor effects were uncommon, but the proportion scoring the maximum of perfect health was high in all scales except for the anxiety/depression dimension and the pain/discomfort dimension (Table 2).

### **Test-Retest Reliability of EQ-5D-5L utility index**

The two-way mixed-effects model, ICC was high for all EQ-5D-5L dimensions and EQ-5D index scores, which lies within the strong agreement range across all respondents ( $ICC > 0.7, p < 0.001$ ) (Table 3).

### **Test-Retest Reliability of EQ-VAS Scores**

The two-way mixed-effects model, ICC for all groups for EQ-VAS was found to be 0.828 (95% CI = 0.775 - 0.866), which indicated good agreement between the two sets of EQ-VAS scores (Table 3).

### **Known Group Validity**

Respondents with mental health disorders, and those who had experience of chronic illness reported statistically significantly more problems on the EQ-5D-5L dimensions of self-care, doing usual activities, having pain or discomfort, feeling anxiety or depression, and EQ-VAS score and utility index than those who have no known mental health disorder, and have no experience of chronic illness (Table 4). There was no significant difference between the mental health disorder patient

group and school sample adolescents in the mobility dimension ( $\chi^2 = 6.655, p = 0.084$ ), but having experience of chronic illness was associated with a difference in scores ( $\chi^2 = 23.81, p < 0.001$ ).

There were statistically significant differences in EQ-5D-5L utility index scores and EQ-VAS scores between individuals with no known mental health condition (school samples) and those with mental health disorders ( $\chi^2 = 207.21, p < 0.001, \chi^2 = 133.24, p < 0.001$ ), and experience of chronic illness ( $\chi^2 = 60.71, p < 0.001$ ). Similar results were observed in all dimensions of EQ-5D-5L, except the mobility dimension (Table 4). The discriminant validity shows individuals with major mental health disorders (schizophrenia, bipolar and depression group reported) significantly more problems with anxiety/depression dimension than the adolescents with no known mental health condition.

There was a significant difference in EQ-VAS score and EQ-5D-5L utility index between school sample and individuals with mental health disorder (depression, schizophrenia, and bipolar) ( $p < 0.001$ ) (Figures 2 and 8), and EQ-5D-5L utility index between individuals with depression and bipolar mental disorder ( $p = 0.045$ ) (Fig 8). There was a significant difference in EQ-VAS score and EQ-5D-5L utility index between different experience of chronic illness like in self and no experience of chronic illness ( $p = < 0.001$ ), in self and the family ( $p = < 0.001$ ), and in self and caring for others ( $p = < 0.001$ ) (Figure 4 and 6).

## **Discussion**

This study aimed to investigate the psychometric properties of the EQ-5D-5L among adolescent population in Ethiopia by evaluating the reliability and validity of this instrument among adolescents

with school samples and adolescents with major mental health disorders. The result shows that the Amharic EQ-5D-5L is easy to fill in and the overall small proportion of missing or inappropriate responses demonstrated the feasibility of administering EQ-5D-5L in an adolescent population group. Low prevalence of severe problems with ceiling effects was observed in the different dimensions of the EQ-5D-5L, which is typical for school sample adolescents. The highest proportion of problems was reported on the ‘having pain or discomfort’ and ‘feeling anxiety/depression’ dimensions for both adolescents with no known mental health disorder and with major mental health disorders. These findings are consistent with other EQ-5D-Y studies (17).

Good levels of reliability were observed in EQ-5D-5L dimensions and utility values as well as the EQ-VAS in both school sample adolescents and those with major mental health disorder respondents which is similar with the previous EQ-5D-Y studies (15,17). These values were somewhat lower on pain/discomfort dimension compared to others dimensions. Participants may not have fully understood the aspects which are covered by pain/discomfort dimension in EQ-5D-5L. On the other hand, it is reasonable to expect that reporting of pain might vary over a few weeks, and a lack of correlation does not necessarily mean that the instrument is not reliable.

In general, the observed ability of the EQ-5D-5L to discriminate between the compared group supports the validity of instruments in all dimensions, except with mobility dimension. This may be due to the large ceiling effects on mobility dimension which does not show a significant difference between the compared group. This is not unexpected given that it was not anticipated that mobility would be a particular problem in a school sample or in participants with a mental health condition.

Known group validity was also observed in this study based on differences between EQ-5D-5L scores of patients with major mental health disorders and healthy adolescents. The difference was particularly more marked in patients with schizophrenia and depression. Patient with schizophrenia and depression experience mental, social and physical impairment which can be associated with the nature and severity of the disease or positive and negative symptoms of the conditions (24). Previous studies of EQ-5D-3L and EQ-5D-5L in adults have shown mixed evidence on the validity in populations with mental health conditions. The validity of the EQ-5D-5L has been demonstrated in patients with anxiety and depression including the ability to discriminate across severity groups and reflect changes in health (25–27). However, neither the EQ-5D-3L or the EQ-5D-5L were able to reflect changes in depression symptoms in individuals who had Type 2 Diabetes who were screened for these symptoms (28). There was limited evidence on EQ-5D-5L for patients with schizophrenia and bipolar although the EQ-5D-5L has been found to be reliable and valid in an adolescent and young adult population with post-traumatic stress disorder (29). Mulhern et al (30) found that EQ-5D-3L was able to discriminate across severity groups in those with depression and anxiety as well as in those with schizophrenia although a latter review (31) cast doubt on the utility of generic health measures like the EQ-5D-3L in patients with schizophrenia. Our study indicates that EQ-5D-5L can discriminate between those with and without schizophrenia and bipolar but further evidence is required.

This study has some limitations. The adult measure and disutility weights were applied to drive utilities for adolescents as an Amharic version was available, EQ-5D-Y may have better content validity for this group. There was no cognitive debriefing so the suitability of the dimensions and

comprehensibility was not tested nor did we examine the responsiveness of the instrument to change as there was no intervention. Therefore, further research is recommended to investigate acceptability and suitability of the EQ-5D-5L compared to the EQ-5D-Y as well as the responsiveness of the measure in this population. The use of different interviewers for the school sample adolescent and the patients with major mental health disorders may also have affected the results. Despite these limitations, this study adds to the knowledgebase of the validity of the Amharic EQ-5D-5L in adolescents. The use of the Amharic EQ-5D-5L in adolescents would offer a cost-effective option compared to developing a new value set for the EQ-5D-Y in this low-resource setting.

## **Conclusions**

This first psychometric study among school samples and adolescents with major mental health disorders in Ethiopia reveals that the Amharic EQ-5D-5L is reliable and valid instrument for the measurement of HRQoL.

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## List of Tables

**Table 1. Background characteristics of the respondents**

| <b>Characteristics</b>               | <b>Study sample n (%)</b> |
|--------------------------------------|---------------------------|
| Age (mean, SD)                       | 15.57 (1.3)               |
| <b>Gender</b>                        |                           |
| Female                               | 188 (37.8)                |
| Male                                 | 306 (62.2)                |
| <b>Type of disease</b>               |                           |
| Healthy                              | 201 (40.4)                |
| Bipolar                              | 99 (19.9)                 |
| Schizophrenia                        | 100 (20.1)                |
| Depression                           | 97 (19.5)                 |
| <b>Education</b>                     |                           |
| No formal education                  | 95 (19.1)                 |
| Primary school                       | 228 (35.0)                |
| Secondary school                     | 174 (25.3)                |
| <b>Experience of serious illness</b> |                           |
| In self                              | 226 (45.5)                |
| In family                            | 78 (15.7)                 |
| In self and in family                | 15 (3.0)                  |
| In caring for others                 | 21 (4.2)                  |
| No                                   | 157 (31.6)                |
| <b>Religion</b>                      |                           |
| Christian                            | 273 (54.9)                |
| Muslim                               | 224 (45.1)                |

**Table 2. Percentages of reported problems in the EQ-5D-5L in adolescent population**

|                         | No known mental health disorders (n = 201) |       | Bipolar disorder (n = 99) |       | Schizophrenia (n = 100) |       | Depression (n= 97) |       |
|-------------------------|--------------------------------------------|-------|---------------------------|-------|-------------------------|-------|--------------------|-------|
|                         | n                                          | %     | n                         | %     | n                       | %     | n                  | %     |
| <b>Mobility</b>         |                                            |       |                           |       |                         |       |                    |       |
| No problems             | 190                                        | 94.50 | 96                        | 97.00 | 91                      | 91.00 | 86                 | 88.70 |
| Slight problems         | 9                                          | 4.50  | 2                         | 2.00  | 6                       | 6.00  | 8                  | 8.25  |
| Moderate problems       | 0                                          | 0.00  | 1                         | 1.00  | 2                       | 2.00  | 3                  | 3.10  |
| Severe problems         | 1                                          | 0.50  | 0                         | 0.00  | 1                       | 1.00  | 0                  | 0.00  |
| Unable/extreme problems | 1                                          | 0.50  | 0                         | 0.00  | 0                       | 0.00  | 0                  | 0.00  |
| Total                   |                                            |       |                           |       |                         |       |                    |       |
| <b>Self-care</b>        |                                            |       |                           |       |                         |       |                    |       |
| No problems             | 193                                        | 96.02 | 54                        | 54.54 | 52                      | 52.00 | 60                 | 61.86 |
| Slight problems         | 6                                          | 3.00  | 21                        | 21.21 | 16                      | 16.00 | 22                 | 22.68 |
| Moderate problems       | 1                                          | 0.50  | 14                        | 14.14 | 20                      | 20.00 | 12                 | 12.37 |
| Severe problems         | 1                                          | 0.50  | 9                         | 9.09  | 9                       | 9.00  | 3                  | 3.09  |
| Extreme problems        | 0                                          | 0.00  | 1                         | 1.01  | 3                       | 3.00  | 0                  | 0.00  |
| Total                   |                                            |       |                           |       |                         |       |                    |       |
| <b>Usual activities</b> |                                            |       |                           |       |                         |       |                    |       |
| No problems             | 166                                        | 82.58 | 39                        | 39.04 | 26                      | 26.00 | 34                 | 35.05 |
| Slight problems         | 29                                         | 14.43 | 13                        | 13.13 | 25                      | 25.00 | 23                 | 23.71 |
| Moderate problems       | 6                                          | 2.99  | 27                        | 27.27 | 26                      | 26.00 | 27                 | 27.84 |
| Severe problems         | 0                                          | 0.00  | 19                        | 19.19 | 18                      | 18.00 | 13                 | 13.40 |
| Extreme problems        | 0                                          | 0.00  | 1                         | 1.01  | 5                       | 5.00  | 0                  | 0.00  |
| Total                   |                                            |       |                           |       |                         |       |                    |       |
| <b>Pain/Discomfort</b>  |                                            |       |                           |       |                         |       |                    |       |
| No problems             | 153                                        | 76.12 | 51                        | 51.51 | 43                      | 43.00 | 34                 | 35.05 |
| Slight problems         | 41                                         | 20.40 | 22                        | 22.22 | 34                      | 34.00 | 39                 | 40.21 |
| Moderate problems       | 3                                          | 1.50  | 23                        | 23.23 | 12                      | 12.00 | 19                 | 19.59 |
| Severe problems         | 2                                          | 0.99  | 2                         | 2.02  | 8                       | 8.00  | 5                  | 5.15  |
| Extreme problems        | 1                                          | 0.49  | 1                         | 1.01  | 3                       | 3.00  | 0                  | 0.00  |
| Total                   |                                            |       |                           |       |                         |       |                    |       |

| <b>Anxiety/Depression</b>                |               |       |               |       |               |       |               |       |
|------------------------------------------|---------------|-------|---------------|-------|---------------|-------|---------------|-------|
| No problems                              | 130           | 64.68 | 30            | 30.30 | 20            | 20.00 | 6             | 6.19  |
| Slight problems                          | 52            | 25.87 | 36            | 36.36 | 41            | 41.00 | 30            | 30.93 |
| Moderate problems                        | 16            | 8.00  | 26            | 26.26 | 21            | 21.00 | 36            | 37.11 |
| Severe problems                          | 2             | 0.99  | 6             | 6.06  | 13            | 13.00 | 24            | 24.74 |
| Extreme problems                         | 1             | 0.49  | 1             | 1.01  | 5             | 5.00  | 1             | 1.03  |
| Total                                    |               |       |               |       |               |       |               |       |
| <b>EQ-VAS score (mean, SD)</b>           | 92.37 (11.81) |       | 72.12 (19.15) |       | 67.55 (19.55) |       | 68.81 (16.05) |       |
| <b>EQ-5D-5L Utility score (mean, SD)</b> | 0.96 (0.094)  |       | 0.83 (0.21)   |       | 0.75 (0.32)   |       | 0.78 (0.20)   |       |
| <b>Minimum EQ-5D-5L Utility score</b>    | 0.27          |       | -0.42         |       | -0.43         |       | 0.20          |       |
| <b>Maximum EQ-5D-5L Utility score</b>    | 1.00          |       | 1.00          |       | 1.00          |       | 1.00          |       |

**Table 3. Agreement between first and second EQ-5D-5L dimension scores**

| EQ-5D-5L scale     | ICC   | ICC range     | P-value |
|--------------------|-------|---------------|---------|
| Mobility           | 0.777 | 0.739 - 0.809 | < 0.001 |
| Self-care          | 0.836 | 0.800 - 0.865 | < 0.001 |
| Usual activity     | 0.790 | 0.724 - 0.838 | < 0.001 |
| Pain/discomfort    | 0.720 | 0.672 - 0.762 | < 0.001 |
| Anxiety/depression | 0.754 | 0.706 - 0.794 | < 0.001 |
| EQ-5D index score  | 0.823 | 0.788- 0.852  | < 0.001 |
| EQ-VAS score       | 0.828 | 0.775- 0.866  | < 0.001 |

**ICC:** Intraclass correlation coefficient

**EQ-5D-5L:** EuroQol five-dimensional three/five-level

**EQ-VAS:** European Quality Visual Analogue Scale

**Statistical analysis:** A two-way mixed-effects model, absolute intraclass correlation coefficient ICC

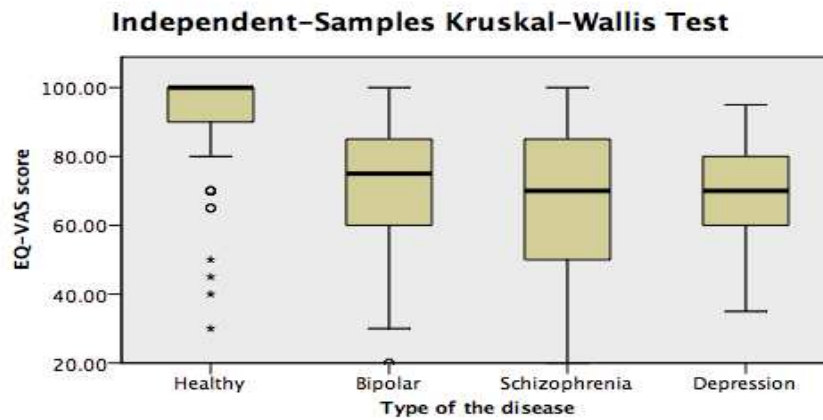
**Table 4. Known group validity on EQ-5D-5L dimension and EQ-VAS scores by health, gender and educational status**

| Dimensions | Disease condition |         |               |            | Chronic illness history |         |           |      |                  | Sex    |        | Educational status |         |           |
|------------|-------------------|---------|---------------|------------|-------------------------|---------|-----------|------|------------------|--------|--------|--------------------|---------|-----------|
|            | No                | Bipolar | Schizophrenia | Depression | No                      | In self | In family | Both | In caring others | Male   | Female | Informal           | Primary | Secondary |
| n          | 201               | 99      | 100           | 97         | 157                     | 226     | 78        | 15   | 21               | 309    | 188    | 95                 | 228     | 174       |
| Mean rank  | 49357             | 23716   | 25447         | 25235      | 42135                   | 54439   | 18340     | 3480 | 5360             | 75379  | 48374  | 23032              | 56866   | 43856     |
| Chi-square | 6.655             |         |               |            | 23.811                  |         |           |      |                  | 5.288  |        | 1.455              |         |           |
| p-value    | 0.0837            |         |               |            | < 0.001                 |         |           |      |                  | 0.0215 |        | 0.4831             |         |           |
| n          | 201               | 99      | 100           | 97         | 157                     | 226     | 78        | 15   | 21               | 309    | 188    | 95                 | 228     | 174       |
| Mean rank  | 38023             | 29062   | 30421         | 2624       | 36765                   | 63061   | 16721     | 3215 | 3992             | 75459  | 48294  | 27824              | 53956   | 41974     |
| Chi-square | 99.022            |         |               |            | 33.063                  |         |           |      |                  | 1.470  |        | 17.851             |         |           |
| p-value    | < 0.001           |         |               |            | < 0.001                 |         |           |      |                  | 0.2254 |        | < 0.001            |         |           |
| n          | 201               | 99      | 100           | 97         | 157                     | 226     | 78        | 15   | 21               | 309    | 188    | 95                 | 228     | 174       |
| Mean rank  | 33474             | 29341   | 32451         | 28488      | 35027                   | 66378   | 15720     | 2950 | 3681             | 75196  | 48557  | 29031              | 52662   | 42061     |
| Chi-square | 136.414           |         |               |            | 51.490                  |         |           |      |                  | 1.510  |        | 22.468             |         |           |
| p-value    | < 0.001           |         |               |            | < 0.001                 |         |           |      |                  | 0.2191 |        | < 0.001            |         |           |
| n          | 201               | 99      | 100           | 97         | 157                     | 226     | 78        | 15   | 21               | 309    | 188    | 95                 | 228     | 174       |
| Mean rank  | 38701             | 26598   | 28644         | 29314      | 38291                   | 60563   | 16848     | 3693 | 3863             | 73766  | 49491  | 26899              | 51360   | 44997     |
| Chi-square | 65.367            |         |               |            | 14.240                  |         |           |      |                  | 4.791  |        | 14.930             |         |           |
| P-value    | < 0.001           |         |               |            | 0.0066                  |         |           |      |                  | 0.0286 |        | 0.0006             |         |           |
| n          | 201               | 99      | 100           | 97         | 157                     | 226     | 78        | 15   | 21               | 309    | 188    | 95                 | 228     | 174       |
| Mean rank  | 33735             | 25947   | 29610         | 34463      | 36176                   | 64611   | 16227     | 3786 | 2955             | 74553  | 49201  | 28891              | 50618   | 44245     |
| Chi-square | 142.246           |         |               |            | 39.541                  |         |           |      |                  | 2.611  |        | 24.585             |         |           |
| P-value    | < 0.001           |         |               |            | < 0.001                 |         |           |      |                  | 0.1061 |        | < 0.001            |         |           |

|                          |           |       |       |       |           |           |           |          |      |           |        |       |         |       |
|--------------------------|-----------|-------|-------|-------|-----------|-----------|-----------|----------|------|-----------|--------|-------|---------|-------|
| EQ-VAS<br>mean<br>score  | 92.4      | 72.1  | 67.5  | 68.8  | 81.4      | 72.8      | 85.3      | 86.0     | 93.9 | 80.1      | 76.6   | 69.4  | 81.8    | 79.9  |
| n                        | 201       | 99    | 100   | 97    | 157       | 226       | 78        | 15       | 21   | 309       | 188    | 95    | 228     | 174   |
| Mean rank                | 7233<br>9 | 19146 | 16467 | 15802 | 4234<br>3 | 4518<br>7 | 2370<br>0 | 466<br>6 | 7859 | 7895<br>9 | 44795  | 17761 | 62387   | 43605 |
| Chi-square               | 207.213   |       |       |       |           | 60.709    |           |          |      |           | 1.719  |       | 24.894  |       |
| P-value                  | 0.0001    |       |       |       |           | 0.0001    |           |          |      |           | 0.1899 |       | < 0.001 |       |
| Utility<br>mean<br>score | 0.96      | 0.84  | 0.76  | 0.78  | 0.85      | 0.83      | 0.92      | .88      | 0.96 | 0.87      | 0.83   | 0.80  | 0.88    | 0.85  |
| n                        | 201       | 99    | 100   | 97    | 157       | 226       | 78        | 15       | 21   | 309       | 188    | 95    | 228     | 174   |
| Mean rank                | 186       | 219   | 186   | 165   | 267       | 209       | 296       | 278      | 344  | 259       | 233    | 193   | 282     | 237   |
| Chi-square               | 133.243   |       |       |       |           | 39.50     |           |          |      |           | 3.906  |       | 28.4362 |       |
| p-value                  | < 0.001   |       |       |       |           | < 0.001   |           |          |      |           | 0.048  |       | < 0.001 |       |

**Statistical analysis:** Non-parametric Kruskal-Wallis rank test

Figure 1. Shows the distribution of EQ-VAS scores between disease group.

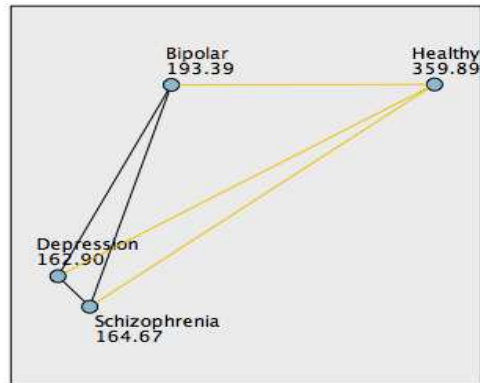


|                                |         |
|--------------------------------|---------|
| Total N                        | 497     |
| Test Statistic                 | 207.213 |
| Degrees of Freedom             | 3       |
| Asymptotic Sig. (2-sided test) | .000    |

1. The test statistic is adjusted for ties.

Figure 2. The pairwise comparison of EQ-VAS scores within the type of disease.

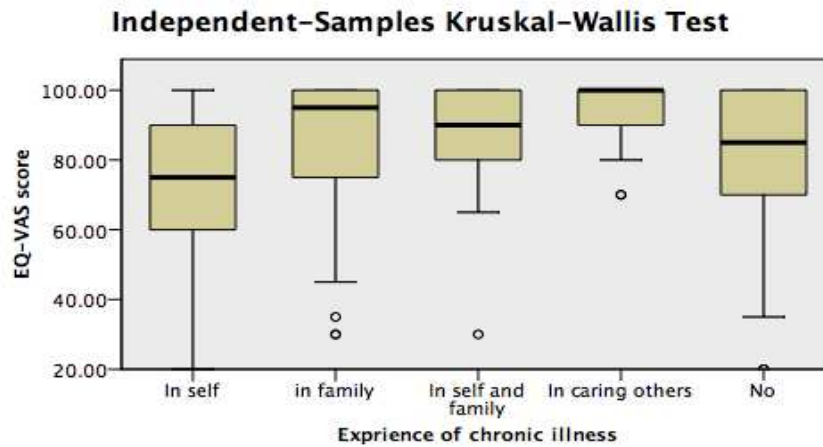
Pairwise Comparisons of Type of the disease



| Sample1-Sample2          | Test Statistic | Std. Error | Std. Test Statistic | Sig. | Adj.Sig. |
|--------------------------|----------------|------------|---------------------|------|----------|
| Depression-Schizophrenia | 1.768          | 20.309     | .087                | .931 | 1.000    |
| Depression-Bipolar       | 30.492         | 20.359     | 1.498               | .134 | .805     |
| Depression-Healthy       | 196.991        | 17.618     | 11.181              | .000 | .000     |
| Schizophrenia-Bipolar    | 28.724         | 20.204     | 1.422               | .155 | .931     |
| Schizophrenia-Healthy    | 195.223        | 17.439     | 11.195              | .000 | .000     |
| Bipolar-Healthy          | 166.499        | 17.498     | 9.516               | .000 | .000     |

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .05.

Figure 3. Shows the distribution of EQ-VAS scores between different experience of chronic illness.

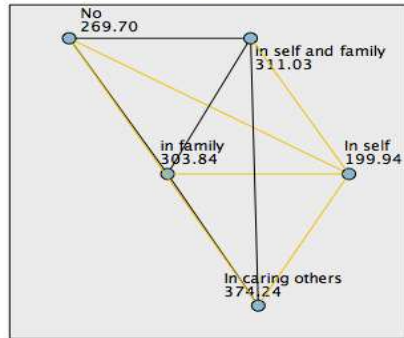


|                                       |        |
|---------------------------------------|--------|
| <b>Total N</b>                        | 497    |
| <b>Test Statistic</b>                 | 60.709 |
| <b>Degrees of Freedom</b>             | 4      |
| <b>Asymptotic Sig. (2-sided test)</b> | .000   |

1. The test statistic is adjusted for ties.

Figure 4. Shows the pairwise comparison of EQ-VAS scores within the different in.

Pairwise Comparisons of Expreience of chronic illness

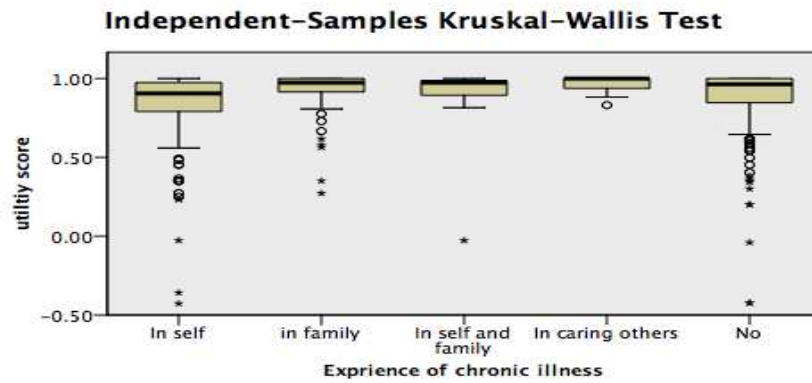


Each node shows the sample average rank of Expreience of chronic illness.

| Sample1-Sample2                     | Test Statistic | Std. Error | Std. Test Statistic | Sig. | Adj.Sig. |
|-------------------------------------|----------------|------------|---------------------|------|----------|
| In self-No                          | -69.757        | 14.806     | -4.712              | .000 | .000     |
| In self-in family                   | -103.899       | 18.714     | -5.552              | .000 | .000     |
| In self-In self and family          | -111.093       | 37.996     | -2.924              | .003 | .035     |
| In self-In caring others            | -174.298       | 32.510     | -5.361              | .000 | .000     |
| No-in family                        | 34.142         | 19.741     | 1.730               | .084 | .837     |
| No-In self and family               | 41.336         | 38.512     | 1.073               | .283 | 1.000    |
| No-In caring others                 | 104.541        | 33.112     | 3.157               | .002 | .016     |
| in family-In self and family        | -7.194         | 40.177     | -.179               | .858 | 1.000    |
| in family-In caring others          | -70.398        | 35.034     | -2.009              | .044 | .445     |
| In self and family-In caring others | -63.205        | 48.176     | -1.312              | .190 | 1.000    |

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .05.

Figure 5. Shows the distribution of utility scores between different experience of chronic illness.

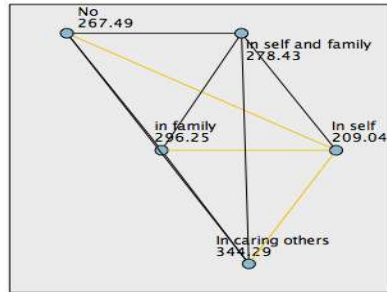


|                                       |        |
|---------------------------------------|--------|
| <b>Total N</b>                        | 497    |
| <b>Test Statistic</b>                 | 39.500 |
| <b>Degrees of Freedom</b>             | 4      |
| <b>Asymptotic Sig. (2-sided test)</b> | .000   |

1. The test statistic is adjusted for ties.

Figure 6. Shows the pairwise comparison of utility scores within the different experience of chronic illness.

Pairwise Comparisons of Expreience of chronic illness

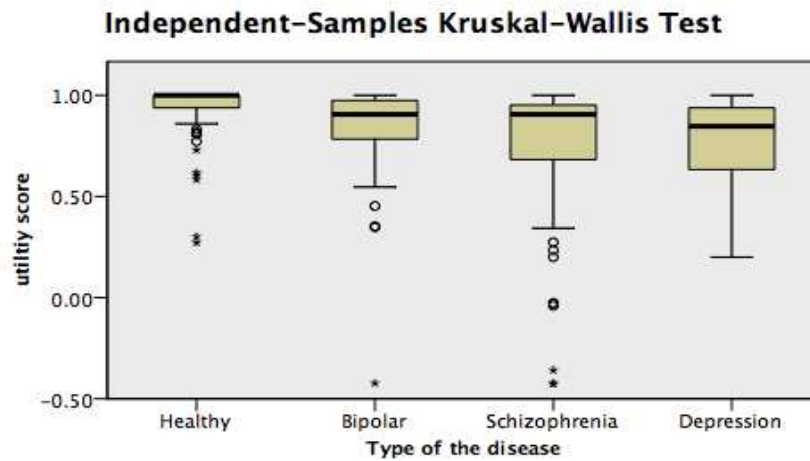


Each node shows the sample average rank of Expreience of chronic illness.

| Sample1-Sample2                     | Test Statistic | Std. Error | Std. Test Statistic | Sig. | Adj.Sig. |
|-------------------------------------|----------------|------------|---------------------|------|----------|
| In self-No                          | -58.445        | 14.714     | -3.972              | .000 | .001     |
| In self-In self and family          | -69.391        | 37.762     | -1.838              | .066 | .661     |
| In self-in family                   | -87.208        | 18.599     | -4.689              | .000 | .000     |
| In self-In caring others            | -135.244       | 32.310     | -4.186              | .000 | .000     |
| No-In self and family               | 10.946         | 38.275     | .286                | .775 | 1.000    |
| No-in family                        | 28.763         | 19.619     | 1.466               | .143 | 1.000    |
| No-In caring others                 | 76.798         | 32.908     | 2.334               | .020 | .196     |
| In self and family-in family        | 17.817         | 39.930     | .446                | .655 | 1.000    |
| In self and family-In caring others | -65.852        | 47.879     | -1.375              | .169 | 1.000    |
| in family-In caring others          | -48.036        | 34.818     | -1.380              | .168 | 1.000    |

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .05.

Figure 7. Shows the distribution of utility scores between type of diseases.

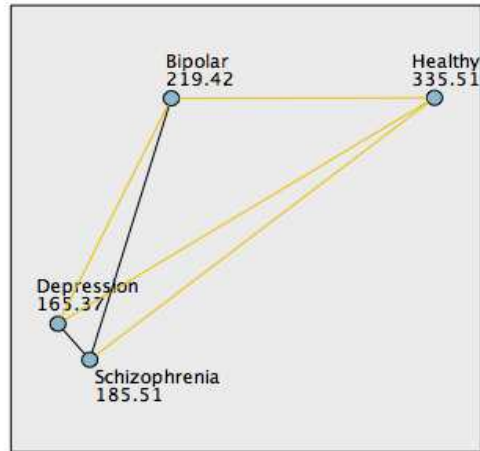


|                                |         |
|--------------------------------|---------|
| Total N                        | 497     |
| Test Statistic                 | 133.243 |
| Degrees of Freedom             | 3       |
| Asymptotic Sig. (2-sided test) | .000    |

1. The test statistic is adjusted for ties.

Figure 8. Shows the pairwise comparison of utility scores within the disease types.

Pairwise Comparisons of Type of the disease



Each node shows the sample average rank of Type of the disease.

| Sample1-Sample2          | Test Statistic | Std. Error | Std. Test Statistic | Sig. | Adj.Sig. |
|--------------------------|----------------|------------|---------------------|------|----------|
| Depression-Schizophrenia | 20.144         | 20.183     | .998                | .318 | 1.000    |
| Depression-Bipolar       | 54.058         | 20.234     | 2.672               | .008 | .045     |
| Depression-Healthy       | 170.149        | 17.509     | 9.718               | .000 | .000     |
| Schizophrenia-Bipolar    | 33.914         | 20.080     | 1.689               | .091 | .547     |
| Schizophrenia-Healthy    | 150.005        | 17.331     | 8.655               | .000 | .000     |
| Bipolar-Healthy          | 116.091        | 17.390     | 6.676               | .000 | .000     |

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .05.